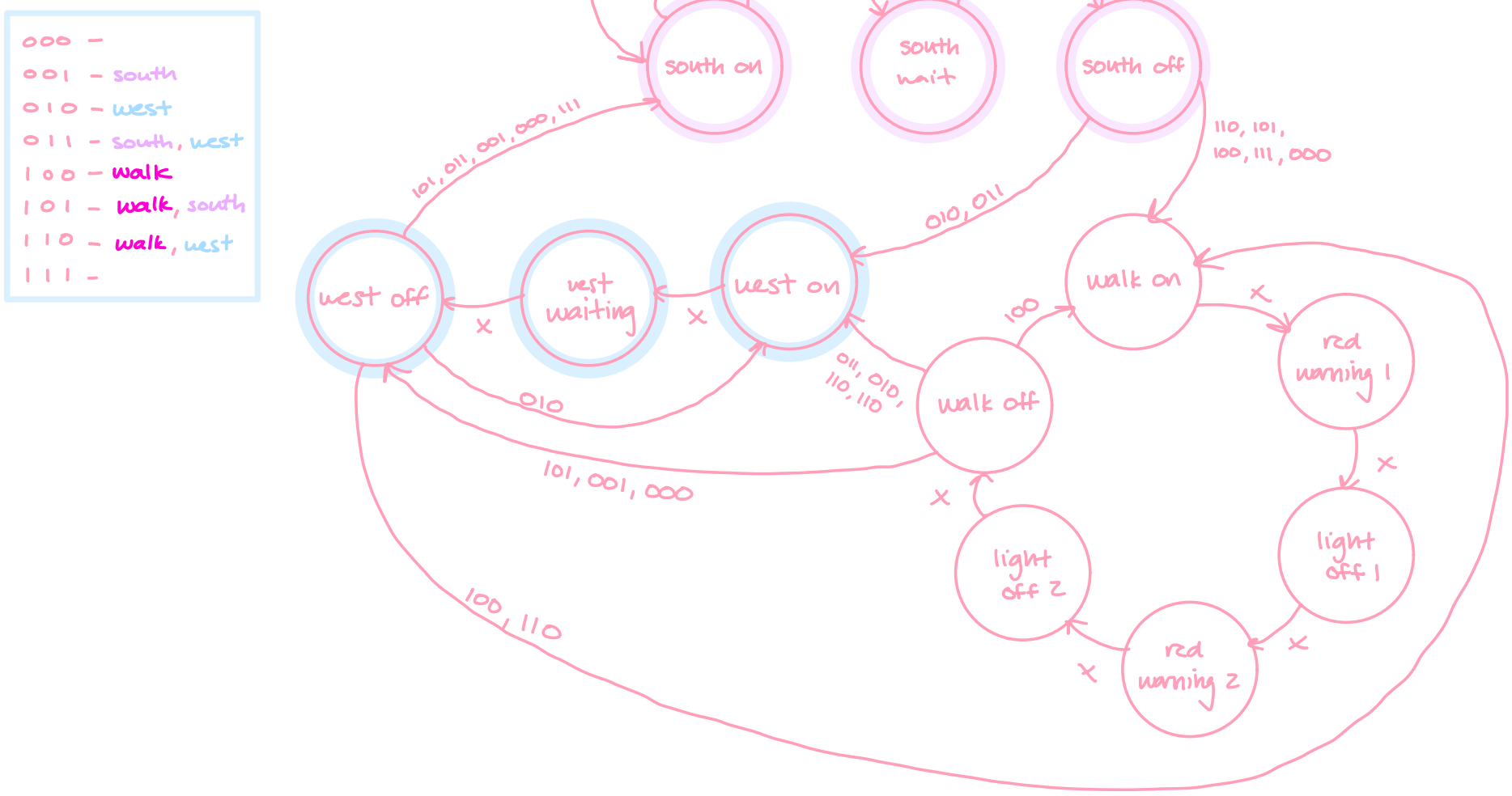
1) Melinda Dai

Dr. Holt

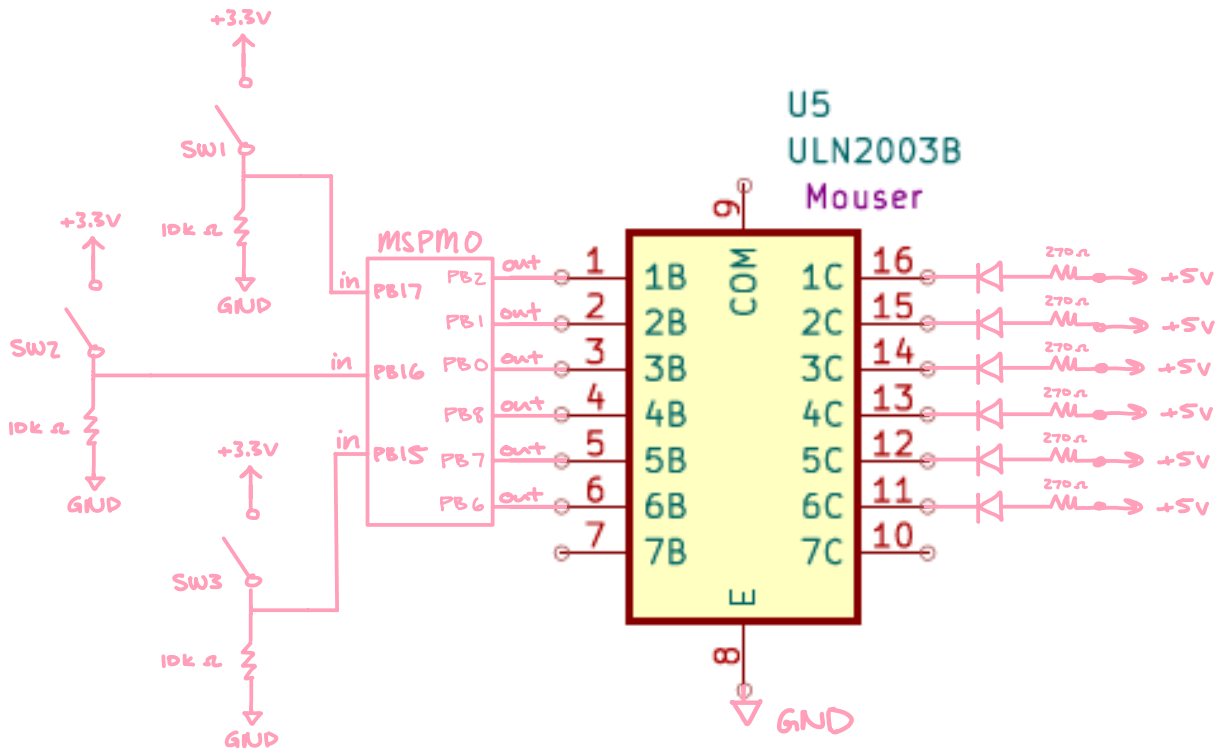
mmd3728

2) SOn: x04000101
SWait: x04000102
SOff: x04000104
WesOn: x04000044
WesWait: x04000084
WesOff: x04000104
WalOn: xOC400104
Wan1: x04000104
WOff1: x00000104
Wan2: x04000104
WOff2: x00000104
WalOff: x04000104



Current State		000	001	010	011	100	101	110	111
0	SOn	SOn	SOn	SWait	SWait	SWait	SWait	SWait	SWait
1	SWait X	SOff	SOff	SOff	SOff	SOff	SOff	SOff	SOff
2	SOff	WalOn	SOn	WesOn	WesOn	WalOn	WalOn	WalOn	WalOn
3	WalOn X	Wan1	Wan1	Wan1	Wan1	Wan1	Wan1	Wan1	Wan1
4	Wan1 X	WOff1	WOff1	WOff1	WOff1	WOff1	WOff1	WOff1	WOff1
5	WOff1 X	Wan2	Wan2	Wan2	Wan2	Wan2	Wan2	Wan2	Wan2
6	Wan2 X	WOff2	WOff2	WOff2	WOff2	WOff2	WOff2	WOff2	WOff2
7	WOff2 X	WalOff	WalOff	WalOff	WalOff	WalOff	WalOff	WalOff	WalOff
8	WalOff	WesOff	WesOff	WesOn	WesOn	WesOn	WesOff	WesOn	WesOn
9	WesOn X	WesWait	WesWait	WesWait	WesWait	WesWait	WesWait	WesWait	WesWait
10	WesWait X	WesOff	WesOff	WesOff	WesOff	WesOff	WesOff	WesOff	WesOff
11	WesOff	SOn	SOn	WesOn	SOn	WalOn	SOn	WalOn	SOn

3)



4)

```
0x20200000  DataBuffer
0x20200000  __start__…
0x20200000  __start__…
0x20200000  __stop___…
0x20200000  __stop___…
0x20200000  0x00000001
0x20200004  0x00000002
0x20200008  0x00000004
0x2020000C  0x00000040
0x20200010  0x00000080
0x20200014  0x00000100
0x20200018  0x0C400000
0x2020001C  0x04000000
```

5)

```
Lab 4, Spring 2026, Step 2. Debug switches
EID1= MMD3728
Switch= 0x1
Switch= 0x0
Switch= 0x2
Switch= 0x0
Switch= 0x4
Switch= 0x0
```

6)

```
0x20200000  DataBuffer __start___llvm_prf_bits __start___llvm_prf_cnts
0x20200000  __stop___llvm_prf_bits __stop___llvm_prf_cnts
0x20200000  0x00040102  0x01040202  0x02040402  0x03040407  0x04040402  0x05040400
0x20200018  0x06040402  0x07040400  0x08040402  0x09010402  0x0A020402  0x0B040402
0x20200030  0x00040102  0x01040202  0x02040402  0x03040407  0x04040402  0x05040400
0x20200048  0x06040402  0x07040400  0x08040402  0x09010402  0x0A020402  0x0B040402
0x20200060  0x00040102  0x01040202  0x02040402  0x03040407  0x04040402  0x05040400
0x20200078  0x06040402  0x07040400  0x08040402  0x09010402  0x0A020402  0x0B040402
0x20200090  0x00040102  0x01040202  0x02040402  0x03040407  0x00000000  0x00000000
0x202000A8  0x00000000  0x00000000  0x00000000  0x00000000  0x00000000  0x00000000
0x202000C0  0x00000000  0x00000000
```

7)

```
Lab 4, Spring 2026
EID1=MMD3728
Option A, South,  connect Red=PB2, Yellow=PB1, Green=PB0
Option X, West,   connect Red=PB8, Yellow=PB7, Green=PB6
Option 0, Sensor, connect Walk=PB17, South=PB16, West=PB15
When all inputs true, ... South, Walk, West, South, Walk, West, South, Walk, West, ... .....
Initialization, Grader started, Activate all three inputs.
Looking for Go South
Wait South
All stop
Go Walk
All stop
Wait Walk
All stop
Wait Walk
All stop
Go West
Wait West
All stop
Go South
Found Go South, Score= 11
Looking for Go Walk
Wait South
All stop
Go Walk
Found Go Walk, Score= 18
Looking for Go West
All stop
Wait Walk
All stop
Wait Walk
All stop
Go West
Found Go West, Finished, Score= 25
```